

Group Theory and Its Applications

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Work in progress. These notes are written up as the course runs; several chapters are still an outline only, and nothing here has been through a careful revision. Corrections are welcome.

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Principal texts

- [Moore] G. W. Moore. *Abstract Group Theory. Physics 618: Applied Group Theory*. Lecture notes, Chapter 1; later chapters dated 2018–2019. 2023. URL: <https://www.physics.rutgers.edu/~gmoore/618Spring2023/GTLect1-AbstractGroupTheory-2023.pdf> (visited on 09/08/2026).
- [Zee] A. Zee. *Group Theory in a Nutshell for Physicists*. Princeton: Princeton University Press, 2016. elementary; a good entry point.
- [Dres] M. S. Dresselhaus, G. Dresselhaus, and A. Jorio. *Group Theory: Application to the Physics of Condensed Matter*. Berlin: Springer, 2008. many worked physical applications.
- [Tao] 陶瑞宝. 物理学中的群论. 高等教育出版社, 2011. a standard Chinese textbook.
- [Ma] 马中骐. 物理学中的群论. 科学出版社, 2006. a standard Chinese textbook.
- [Chen] 陈金全. 群表示论的新途径. 上海: 上海科学技术出版社, 1984. Nanjing University.